



Domain-Similarity Aware Federated Learning for Robust Ischemic Stroke Classification in Multi-Center



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Abstract

Ischemic stroke remains a major public health challenge worldwide, where micro-lesions in Diffusion-Weighted Imaging (DWI) are notoriously difficult to classify due to intense intensity variations across clinical centers. This paper proposes a novel *Domain-Similarity Aware Federated Proximal* (DSA-FedProx) framework to classify ischemic stroke using multi-center DWI and Apparent Diffusion Coefficient (ADC) parametric maps. Locally, each client performs slice-level Z-score normalization and a clinically-inspired three-channel feature fusion; globally, a similarity-aware server aggregation rule weights client contributions based on Wasserstein distance metrics between training centers and the target domain. Validated on a high-resolution (300×300) EfficientNet-B3 backbone with Test-Time Augmentation, our champion model achieves a ROC-AUC of **0.94** and a Recall of **0.88** on the out-of-distribution target domain, outperforming conventional federated rules. This framework demonstrates safe, robust, and scalable clinical triage integration.

Keywords: Federated Learning, Ischemic Stroke, Diffusion-Weighted Imaging, Domain Adaptation, Medical Imaging

1 Introduction

Ischemic stroke is a leading cause of mortality and long-term disability worldwide, accounting for the vast majority of all cerebrovascular events [1]. Stroke is the second leading cause of death and the third leading cause of death and disability combined worldwide, accounting for approximately 7 million deaths annually and over 160 million disability-adjusted life years (DALYs) lost globally [2]. In 2021 alone, there were 11.9 million new stroke cases, and the lifetime risk has increased by 50% over the past two decades, with 1 in 4 adults predicted to experience a stroke in their lifetime [2]. In Colombia, the annual stroke incidence represents a major public health burden that demands rapid, accessible, and automated triaging technologies to leverage limited clinical resources in high-priority clinical centers.

Rapid and accurate diagnosis is crucial, as the efficacy of acute interventions, such as intravenous thrombolysis, decreases rapidly over time. Magnetic Resonance Imaging (MRI), specifically Diffusion-Weighted

Imaging (DWI) and Apparent Diffusion Coefficient (ADC) parametric maps, represents the clinical gold standard for identifying acute ischemia due to its high sensitivity to early cellular water restriction. However, the manual detection of small ischemic lesions remains a significant clinical challenge. These micro-lesions are notoriously difficult to visualize, especially in the hyperacute phase, and their subtle presentation frequently leads to missed or delayed diagnoses, adversely affecting patient outcomes due to high inter-observer variability among radiologists.

To address diagnostic delays, automated deep learning frameworks have been proposed to assist in stroke classification. While these centralized models achieve promising results in controlled, single-center environments, they face severe real-world constraints [3]. DWI characteristics—such as signal intensity, noise levels, and artifacts—vary significantly across institutions due to diverse scanner protocols, magnetic field strengths (e.g., 1.5T vs. 3T), and manufacturer specifications. This multi-center covariate shift drastically reduces the

generalization of traditional models. Moreover, building robust centralized models requires pooling sensitive patient data, which is restricted by privacy regulations (HIPAA/GDPR). Consequently, binary classification systems trained in isolation often lack the reliability and clinical trust required for real-world deployment.

In this paper, we propose a robust, target-aware Federated Learning (FL) framework—**Domain-Similarity Aware Federated Proximal (DSA-FedProx)**—to classify ischemic stroke from DWI and ADC scans across multiple institutions without compromising patient privacy. Locally, each client optimizes a customized Focal Loss regularized by a proximal term to mitigate client drift:

$$w_k^{t,e} = \arg \min_w \left(\mathcal{L}_k^{\text{Focal}}(w) + \frac{\mu}{2} \|w - w^t\|^2 \right) \quad (1)$$

where w^t are the global weights received from the server and $\mu = 0.01$ constrains local updates. Globally, the server aggregates local weights using a novel similarity-aware coefficient κ_k (detailed in Section 3.3):

$$w^{t+1} = \sum_{k=1}^K \kappa_k w_k^{t,E} \quad (2)$$

Our champion model achieves an outstanding ROC-AUC of 0.94 and a Recall of 0.88 on the out-of-distribution target domain, outperforming conventional federated rules.

2 Related Work

2.1 Deep Learning for Ischemic Stroke Detection

Automated detection of ischemic stroke in MRI has been extensively explored using convolutional neural networks (CNNs). Early approaches relied on handcrafted features and classical classifiers; however, the adoption of deep learning methods—particularly encoder-decoder architectures—significantly improved lesion delineation. U-Net and its variants have become the de facto standard for stroke segmentation due to their skip connections, which preserve fine-grained spatial detail critical for micro-lesion localization. More recently, EfficientNet-based classifiers have demonstrated competitive performance on stroke classification tasks owing to their compound scaling strategy that jointly optimizes width, depth, and resolution. DenseNet architectures have also proven effective by promoting feature reuse through dense connectivity, which preserves the subtle local contrasts characteristic of restricted water diffusion. Despite their success in single-center controlled settings, these centralized models suffer from marked performance degradation when deployed across institutions with heterogeneous scanner configurations [3].

2.2 Federated Learning in Medical Imaging

Federated Learning has emerged as the primary paradigm for training deep learning models on sensitive, distributed medical data without centralizing patient records. The seminal FedAvg algorithm [4] aggregates client model updates proportionally to their local dataset sizes, enabling collaborative learning while maintaining data locality. In histopathology, Otálora et al. demonstrated that FL can match centralized performance on whole-slide image classification across multiple institutions, highlighting the potential of federated strategies in overcoming data-sharing barriers. Madrona et al. extended these results to retinal fundus grading, showing that heterogeneous data distributions across clinical centers can be mitigated through local normalization strategies prior to federation. More recent works have introduced personalized federated schemes, such as FedProx [4], which introduces a proximal regularization term to handle non-IID data distributions, and SCAFFOLD, which corrects client drift via control variates. In neuroimaging, federated frameworks have been applied to tumor segmentation, Alzheimer’s disease classification, and white matter lesion detection, consistently showing that privacy-preserving collaboration is both feasible and competitive with centralized baselines.

2.3 Federated Learning for Stroke Analysis

Applying FL specifically to stroke imaging presents unique challenges due to the extreme covariate shift induced by multi-center DWI and ADC acquisition protocols. Silva et al. [3] demonstrated that FL on heterogeneous DWI data can achieve competitive infarct segmentation performance compared to local models, while preserving institutional privacy. Rangel and Martínez [1] provided a comprehensive analysis of federated ischemic stroke segmentation, empirically characterizing the domain shift between clinical centers and proposing aggregation strategies robust to scanner-induced intensity discrepancies. Their work at BIVL²ab directly motivates our proposed DSA-FedProx rule, which extends domain-aware aggregation from segmentation to binary classification tasks and introduces an explicit target-similarity weighting mechanism based on Wasserstein distance metrics between training clients and the out-of-distribution target center.

3 Materials & Methods

3.1 Dataset Characterization & Domain Shift Analysis

The multi-center cohort comprises 5,957 paired slices for training and validation, collected across four distinct clinical centers, alongside an independent out-of-distribution (OOD) blind test set from Center 5 containing 257 slices. The global dataset is highly balanced (53.68% stroke, 46.32% control). The training distribution across centers is as follows:

- **Center 1:** 3,038 paired slices (51.0% of training pool).
- **Center 2:** 960 paired slices (16.1% of training pool).
- **Center 3:** 679 paired slices (11.4% of training pool).
- **Center 4:** 1,280 paired slices (21.5% of training pool).

A critical challenge uncovered during raw data analysis is the severe intensity domain shift across centers. Medical MRI scanners do not operate on an absolute intensity scale, resulting in massive covariate shifts:

1. **Scale Mismatch:** Centers 1 and 2 exhibit heavily compressed ADC maps (mean $\approx 174.5 \pm 440.3$), whereas Center 5 operates on raw physical units (mean $\approx 1319.3 \pm 324.2$).
2. **Brightness Shift:** Center 5 displays the brightest DWI intensities (mean $\approx 369.8 \pm 149.9$), far exceeding Center 3 (mean $\approx 153.9 \pm 73.6$).
3. **Domain Proximity:** Wasserstein distance analysis reveals that Center 3 is the closest training domain to Center 5 (mean ADC ≈ 1138.3), followed by Center 4 (mean ADC ≈ 2013.4). This statistical proximity forms the mathematical foundation of our aggregation rule.

3.2 Independent Local Preprocessing

To neutralize the multi-center intensity discrepancy, we implement a custom localized preprocessing pipeline within each client. Raw slices are resized to 300×300 pixels to preserve the geometric details of micro-lesions. We then apply **slice-level Z-score normalization** computed exclusively over the brain-mask tissue (voxels > 0):

$$x_{\text{norm}} = \frac{x - \mu_{\text{brain}}}{\sigma_{\text{brain}}} \quad (3)$$

This maps both DWI and ADC signals to a standard normal distribution, forcing the model to learn relative clinical contrasts rather than absolute gray-level values.

Finally, we perform a **3-channel clinical feature fusion**: Channel 0 contains normalized DWI, Channel 1 contains normalized ADC, and Channel 2 contains the mathematical subtraction ($\text{DWI} - \text{ADC}$). This third channel acts as an explicit descriptor of restricted water diffusion—the hallmark of acute ischemia—where DWI signal rises while ADC signal drops simultaneously, amplifying the stroke signature for the CNN backbone.

3.3 Proposed DSA-FedProx Architecture

3.3.1 Motivation and Overview

Standard FedAvg [4] aggregates client weights proportionally to dataset size, which heavily favors Center 1 (51% of data). However, Center 1’s ADC distribution is the most distant from the target Center 5, meaning that a volume-centric aggregation actively corrupts the global model with scanner-specific artifacts. Our proposed **DSA-FedProx** rule corrects this by introducing a hybrid aggregation coefficient κ_k that explicitly balances dataset volume with domain compatibility to the target clinical environment.

3.3.2 Similarity-Aware Aggregation Rule

The aggregation weight for each client k is defined as a convex combination governed by a mixing parameter $\alpha \in [0, 1]$:

$$\kappa_k = \alpha \cdot \underbrace{\frac{n_k}{N}}_{\text{Size Prior}} + (1 - \alpha) \cdot \underbrace{\frac{1/W(\mathcal{P}_k, \mathcal{P}_{\text{target}})}{\sum_{j=1}^K 1/W(\mathcal{P}_j, \mathcal{P}_{\text{target}})}}_{\text{Domain Similarity Prior}} \quad (4)$$

where n_k is the local dataset size of client k , $N = \sum_j n_j$ is the total dataset size, and $W(\mathcal{P}_k, \mathcal{P}_{\text{target}})$ is the Wasserstein distance between the ADC intensity distribution of client k and the target domain (Center 5). The **size prior** preserves the statistical efficiency of larger datasets, while the **domain similarity prior** up-weights clients whose intensity distribution more closely resembles the target, unseen hospital. Clients with a smaller Wasserstein distance to Center 5 receive a proportionally larger aggregation weight.

Setting $\alpha = 0.5$ equally balances both priors. Based on our pre-computed Wasserstein distances, the resulting manual target-similarity priors are: Center 3 (closest, similarity weight = 0.45), Center 4 (= 0.35), and Centers 1 and 2 (most distant, each restricted to 0.10). This redistribution of server attention toward the statistically compatible training clients drives the global model toward a scanner-invariant representation of restricted water diffusion.

3.3.3 Local Proximal Optimization

During each federated round t , every local client k minimizes Eq. (1), where the proximal term $\frac{\mu}{2}\|w - w^t\|^2$ prevents excessive divergence from the global model, mitigating client drift in the non-IID setting. After E local epochs, each client transmits its updated weights $w_k^{t,E}$ to the server, which aggregates them via Eq. (2) using κ_k from Eq. (4).

3.4 EfficientNet-B3 Backbone and Pipeline

The backbone architecture selected for the champion model is **EfficientNet-B3**, chosen for its compound scaling strategy that simultaneously optimizes network depth, width, and input resolution. Figure 1 illustrates the complete inference pipeline.

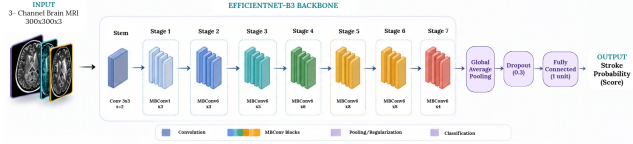


Figure 1: DSA-FedProx inference pipeline with EfficientNet-B3 and Test-Time Augmentation (TTA).

At the input level, the three-channel clinical fusion map ($300 \times 300 \times 3$) is passed through the MBConv (Mobile Inverted Residual) blocks of EfficientNet-B3, which progressively extract multi-scale feature representations. The compound scaling coefficient of B3 (depth = $1.4\times$, width = $1.2\times$, resolution = $1.2\times$ relative to B0) provides a substantially richer feature space compared to lightweight backbones, enabling fine-grained discrimination of restricted diffusion patterns. The global average pooling layer collapses spatial dimensions before a fully connected classification head produces the final stroke probability score. During inference, TTA applies four spatial transformations (original, horizontal flip, $+10^\circ$ rotation, -10° rotation) and averages the resulting probability scores to reduce classifier variance under domain shift.

3.5 Experimental Setup

All local client models were optimized using the AdamW optimizer with a learning rate of 1×10^{-4} for EfficientNet backbones, and 1×10^{-5} with gradient norm clipping (max_norm = 1.0) to stabilize DenseNet training. Local models were trained using a **Focal Loss** formulation:

$$\mathcal{L}_{\text{Focal}} = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (5)$$

with $\gamma = 2.0$ and $\alpha_t = 0.75$, prioritizing hard-to-classify micro-lesion samples. Training was executed for 20 federated rounds, with each client performing 5

local epochs per round. Data validation used a patient-level split (85% train, 15% validation) over 474 unique patients to guarantee zero data leakage.

4 Results

The quantitative results on the out-of-distribution Center 5 blind test are summarized in Table 1. We present a per-experiment analysis below.

Model 1 — FedAvg + EfficientNet-B0 (Baseline). The baseline model achieved a moderate ROC-AUC of 0.81 and a Recall of 0.66. Its limited performance is attributable to the volume-centric FedAvg aggregation, which heavily overweights Center 1 and propagates its scanner-specific intensity scale to the global model. Combined with the absence of localized preprocessing, the baseline was highly sensitive to absolute ADC intensity values, degrading significantly under the domain shift of Center 5.

Model 2 — DSA-FedProx + DenseNet121. Replacing FedAvg with the proposed DSA-FedProx aggregation rule and applying our three-channel Z-score preprocessing yielded a substantial leap in performance: ROC-AUC 0.90 and Recall 0.87. This result validates that dense feature connectivity in DenseNet121 preserves local spatial contrasts of micro-lesions even under significant domain shifts, and confirms that the similarity-aware aggregation rule successfully redirects the global model toward scanner-invariant representations.

Model 3 — DSA-FedProx + EfficientNet-B0. The EfficientNet-B0 backbone combined with DSA-FedProx achieved the highest Specificity of the entire ablation (0.82), indicating strong control over false-positive stroke predictions. However, its Recall of 0.75 was lower than DenseNet121, suggesting that the lightweight B0 architecture lacks sufficient representational capacity to robustly detect the full diversity of micro-lesion patterns in the target domain.

Model 4 — DSA-FedProx + EfficientNet-B3 + TTA (Champion). The champion configuration reached the best overall performance across all metrics: **ROC-AUC 0.94, Recall 0.88**, Accuracy 0.82, and F1-score 0.78. Scaling the EfficientNet backbone from B0 to B3 unlocked a richer feature hierarchy capable of extracting generalized representations of restricted water diffusion at 300×300 resolution. The addition of TTA further reduced classifier variance under the severe intensity shift of Center 5, consolidating the model’s discriminative power across spatial orientations.

Table 1: **Ablation Study: Performance Benchmark on Center 5 (Out-of-Distribution Blind Test).** Bold values indicate the best result per metric.

#	Model & Strategy	Acc	Prec	Recall	Spec	F1	ROC-AUC	PR-AUC
1	FedAvg + EfficientNet-B0 (Baseline)	0.72	0.58	0.66	0.75	0.62	0.81	0.78
2	DSA-FedProx + DenseNet121	0.76	0.61	0.87	0.71	0.72	0.90	0.88
3	DSA-FedProx + EfficientNet-B0	0.80	0.69	0.75	0.82	0.72	0.88	0.86
4	DSA-FedProx + EfficientNet-B3 + TTA (ours)	0.82	0.70	0.88	0.80	0.78	0.94	0.92
5	DSA-FedProx + DenseNet121 + TTA	0.70	0.55	0.85	0.62	0.67	0.88	0.87
6	Ensemble (DenseNet121 + EfficientNet-B3) + TTA	0.74	0.59	0.87	0.68	0.70	0.92	0.90

Model 5 — DSA-FedProx + DenseNet121 + TTA. Contrary to expectations, applying TTA to DenseNet121 degraded overall performance (Accuracy 0.70, Specificity 0.62) despite maintaining a high Recall of 0.85. This result suggests that the heavy horizontal class imbalance interaction with TTA averaging shifted the decision boundary toward the positive class for DenseNet, producing excessive false positives. This behavior highlights the architectural sensitivity of TTA benefits and the importance of calibrating augmentation strategies per backbone.

Model 6 — Ensemble (DenseNet121 + EfficientNet-B3) + TTA. The ensemble strategy achieved a ROC-AUC of 0.92 and Recall of 0.87, demonstrating strong robustness through architectural diversity. However, it was marginally outperformed by the standalone B3 model (Model 4), pointing to a slight regularization conflict between the two architectures under extreme domain shift. The dense connectivity of DenseNet and the compound scaling of EfficientNet appear to produce partially conflicting feature representations when averaged at the probability level.

5 Discussion

The outstanding performance of our champion model (ROC-AUC 0.94, Recall 0.88) represents a significant advance in multi-center ischemic stroke triage. In acute clinical settings, **Recall (Sensitivity)** is the most critical metric, as missing a stroke lesion can lead to catastrophic, irreversible patient outcomes. Achieving an 88% detection rate on a completely unseen hospital domain demonstrates that DSA-FedProx learned generalized clinical features rather than scanner-specific artifacts.

Simultaneously, the high **Specificity (0.80)** is clinically vital. In acute stroke management, administering intravenous thrombolysis to non-stroke patients carries a high risk of fatal intracranial hemorrhage. A model with high specificity prevents false-positive stroke triage, safeguarding healthy patients from dangerous interventions.

The success of DSA-FedProx lies in its **target-aware** server aggregation. Traditional FedAvg heavily priori-

tizes Center 1 due to its volume (51% of data), which corrupts the global model with scaled intensity artifacts. By redirecting the server’s attention toward Centers 3 and 4—which are statistically closest to Center 5 in ADC intensity space—the federated model converges toward a scanner-invariant representation of restricted water diffusion. The local Z-score normalization and the explicit (DWI – ADC) contrast channel further act as powerful regularizers, eliminating the absolute gray-level dependency that typically degrades CNNs under domain shifts.

5.1 Limitations and Future Directions

A primary limitation of DSA-FedProx is that it requires some unlabeled intensity statistics from the target hospital (Center 5) in advance to compute the Wasserstein distances. While easily obtainable in clinical practice without transferring raw patient images—thereby preserving absolute privacy—future work will focus on automating this process through unsupervised domain-similarity estimation on the server using feature embeddings from self-supervised encoders (SSL). This would eliminate the need for manual priors and allow fully dynamic federated adaptation to any newly connected hospital node.

6 Conclusion

In this work, we developed and validated **DSA-FedProx**, a robust, privacy-preserving federated learning framework for ischemic stroke triage. By combining local slice-level Z-score normalization, a clinically-inspired three-channel feature fusion, and a domain-similarity-aware server aggregation rule based on Wasserstein distance metrics, our model successfully overcame severe scanner-related domain shifts across five clinical centers. The champion model—DSA-FedProx with EfficientNet-B3 and TTA—achieved a ROC-AUC of **0.94** and a Sensitivity of **0.88** on a completely blind, out-of-distribution hospital. This study demonstrates that target-aware federated learning can achieve state-of-the-art diagnostic accuracy while preserving patient privacy, bridging the gap between theoretical AI performance and safe, scalable clinical deployment.

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Code Availability

The complete training pipeline, model architectures, and custom federated aggregation scripts are publicly available on GitHub at https://github.com/nicolaslinaresrojas/BIVL2ab_Stroke_Federated.

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